

In the folder you can see draft IPAC26 paper. Could you please independently verify the conclusions and numbers in the first two paragraph of the section "Cost estimations" including Table 4? Depending on what you will find, I will probably replace the exact numbers in Table 4 with ranges, and also will write that two different AI came with certain percentages of each other for the cost estimates. The needed LHeC files with cost estimates and LHeC ESPP submissions, referenced in the paper, are also in the folder.

Here is the verification, point by point.

1. Internal arithmetic of Table 4 — internally consistent, but with a small input slip

The line-to-line additions and the 60 % contingency multiplication in Table 4 are self-consistent ($722.2 + 180.5 = 902.7$ ✓; $902.7 \times 1.6 = 1444.3$ ✓). But the conversions from the paragraph values (578 MCHF, 601 PY) are slightly off:

Line	Paper	Recalc from 578 / 601	Δ
Material 578×1.25	722.2	722.5	−0.3
Labor 601×0.300	180.5	180.3	+0.2
Pre-cont total	902.7	902.8	−0.1
60 % contingency	541.6	541.7	−0.1
Total	1444.3	1444.5	−0.2

So the AI rounded twice in inconsistent directions; the bottom-line error is 0.2 MUSD — negligible, but the conversion arithmetic is technically wrong. If you keep round numbers in Table 4 you should either fix to 722.5 / 180.3 / 902.8 / 541.7 / 1444.5, or — better given your plan — convert it to a range.

2. Sub-totals in paragraph 1 — consistent

Sums are fine: $278 + 50 + 250 = 578$ MCHF ✓, $451 + 50 + 100 = 601$ PY ✓. The 70 MCHF arc cost matches what the LHeC ESPP submission states ("an overall budget reduction of at least 70 MCHF" by removing arcs 3–6 from the 50 GeV multi-pass version), so this number is well grounded.

3. Cross-check of the cost magnitudes against the LHeC CDR scaling

I scaled the LHeC CDR table (CERN-ACC-2018-0061, 60 GeV, 1756 MCHF) to a 6 GeV recirculating linac, 2 passes for 12 GeV, using three scaling scenarios (aggressive / moderate / conservative). Final-IR and detector kept at the paper's manual values (50 + 250 MCHF):

Scenario	Electron complex	Grand total	Ratio vs paper 578	Equivalent USD (60 % cont)
Aggressive (most items dropped/scaled hard)	445	745	1.29×	1777
Moderate	522	822	1.42×	1932
Conservative	645	945	1.63×	2177
Paper	278	578	1.00×	1444

Even the most aggressive linear scaling I could justify from the LHeC CDR sits **~30 % above** the paper. The "278 MCHF electron complex" looks low: it appears to capture roughly SRF + R&D + injector + arcs alone (≈ 380 MCHF on my numbers, or possibly less if SRF was scaled per cavity rather than per metre), with civil engineering, cryo, general/installation, and source items either omitted or heavily compressed. This is consistent with the paper's own caveat that "IR and detector line items were manually increased above the raw AI output" — the raw AI clearly under-counted the auxiliaries on the electron complex side too.

4. The "halves SRF, civil and cryo" wording is a bit aspirational

Going from a single-pass 10 GeV linac to a 6 GeV recirculating linac scales the dominant items by $6/10 = 0.6$, not 0.5. The 70 MCHF arc penalty then leaves a net saving of ~ 62 MCHF, so the *bottom line* of the comparison is reproducible — but the verb "halves" is not literally true. "Reduces by ~ 40 %" would be more accurate.

5. CHF/USD rate and CD-0 vs CD-1

- 1 CHF $\approx$ 1.27 USD as of mid-April 2026; the paper's 1.25 is conservative and within the recent range.
- The "\$1.7–\$2.8 B" range is the **CD-1** range (June 2021), not CD-0 (December 2019, mission-need only). Half the upper bound is exactly \$1.4 B, so the comparison number is right; the label "CD0-approved" is wrong and should read "CD-1".

6. Summary — what to do in the paper

The qualitative conclusions hold up: the hybrid sits well below the EIC CD-1 upper bound regardless of which AI does the scaling. The exact numbers do not.

Plausible ranges to use in Table 4 (rounding to 2 sig fig), built from the spread between the paper's AI scaling and my independent re-scaling:

- Electron complex: **280 – 600 MCHF**
- IR: ~ 50 MCHF (manually set anyway)

- Detector: ~250 MCHF (manually set anyway)
- Pre-contingency MCHF total: **580 – 900 MCHF**
- Pre-contingency MUSD: **900 – 1300 MUSD**
- Total MUSD with 60 % contingency: **~1.4 – 2.1 BUSD**, i.e. ~50 – 75 % of the EIC CD-1 upper bound.

For the "two AIs agree to within X %" wording, the closest defensible number is:

Two independent AI-assisted scalings of the LHeC CDR cost agree on the bottom line of the hybrid EIC-LHeC project total to within roughly $\pm 30\text{--}40\%$ at the pre-contingency level, both lying well below the upper end of the EIC CD-1 cost range.

Suggested specific corrections regardless of whether you switch to ranges:

- replace "CD0-approved" → "CD-1-approved" (cost ranges are set at CD-1, not CD-0);
- replace "halves the dominant SRF, civil, and cryo costs" → "reduces the dominant SRF, civil, and cryo costs by ~40 %";
- in Table 4, either keep 722.5 / 180.3 / 902.8 / 541.7 / 1444.5, or convert the table to ranges as you planned (recommended given the order-of-magnitude caveat already in the text).

Sources:

- [Electron-Ion Collider Achieves Critical Decision 1 Approval \(BNL\)](#)
- [Swiss Franc to USD Spot Exchange Rate \(FRED\)](#)

Thank you! I would like to use ranges in the updated Table 4, and use numbers from your aggressive estimate (because of the large 60% contingency added on top). Could you please clarify how you estimated labor cost (in PY - person years) for the table 4, and what ranges would you suggest to include for each line?

How I estimated labor (PY)

Honest answer: I did **not** independently derive the labor estimate from the LHeC CDR. The CERN-ACC-2018-0061 cost report does not contain person-year estimates — it explicitly says it "does not comment on the required manpower for the facility operation." So when I produced the cross-check, I simply used 600 PY, anchored to the paper's own value (601 PY).

Looking at the paper's implicit assumption, labor was scaled roughly proportionally to material cost:

Sub-system	MCHF	PY	PY / MCHF
Electron complex	278	451	1.62

Sub-system	MCHF	PY	PY / MCHF
IR	50	50	1.00
Detector	250	100	0.40
Total	578	601	1.04

That ~1 PY per MCHF rule of thumb (i.e. labor $\approx$ 25 % of material cost in USD with the paper's conversion factors) is consistent with typical DOE/CERN large-project planning, so it is a defensible scaling. To make the range honest, I'll scale PY by the same factor my aggressive estimate increases material — that way the labor line tracks the harder-data-driven material line.

Suggested ranges for Table 4

Lower bound = paper's existing AI-derived numbers; upper bound = my aggressive LHeC-CDR scaling. Labor scaled by the paper's own 1.04 PY/MCHF heuristic. Conversion factors (1.25 USD/CHF, 300 k\$ /PY, 60 % contingency) kept fixed.

Paragraph 1 (sub-totals leading into the table):

Item	Lower (paper)	Upper (aggressive)
Electron complex	280 MCHF (450 PY)	450 MCHF (720 PY)
IR	50 MCHF (50 PY)	50 MCHF (50 PY)
Detector	250 MCHF (100 PY)	250 MCHF (100 PY)
Grand total	~580 MCHF (~600 PY)	~750 MCHF (~870 PY)

(The IR and detector entries are unchanged because they were already manually set in the paper above raw AI output and don't depend on the LHeC scaling.)

Updated Table 4 with ranges:

Item	Range (MUSD)	How computed
Material costs (580–750 MCHF $\times$ 1.25)	725 – 935	lower from 580×1.25 , upper from 750×1.25
Labor costs (600–870 PY $\times$ 300 k USD)	180 – 260	
Pre-contingency total	905 – 1195	
Contingency (60 %)	545 – 720	
Total with contingency	~1450 – 1915	i.e. ~1.4 – 1.9 B USD

Rounding to one decimal of B USD: **~1.4 – 1.9 B USD**, which the paper can describe as "comfortably below the upper bound of the EIC CD-1 cost range of \$1.7–2.8 B" (and replace "CD0" with "CD-1" while you're in there — CD-0 only sets Mission Need; the cost range was set at CD-1 in June 2021).

Suggested caption update for Table 4: something like *"Hybrid EIC cost range, estimated by two independent AI-assisted scalings of the LHeC CDR cost. The two estimates agree to within ~30 % at the pre-contingency level."*

Sources:

- [LHeC Cost Estimate, CERN-ACC-2018-0061](#) (provided in folder)
- [EIC Critical Decision 1 Approval — BNL Newsroom](#)

Thank you, this is good. Could you please also produce a summary of this dialogue and evaluation that I can share with colleagues (I will post it to GitHub with help from your colleague Claude Code).

I've written a self-contained markdown summary suitable for posting to GitHub.

[View the verification summary](#)

The document covers: the scope and method, the four findings (Table 4 arithmetic, paragraph-1 sub-totals, the LHeC-CDR scaling cross-check with three scenarios, the "halves" overstatement, the CHF/USD rate, and the CD-0 vs CD-1 nomenclature), an explicit note that the LHeC CDR contains no PY estimates and the labor figure is therefore anchored on the paper's own ~1 PY/MCHF heuristic, the recommended wording corrections, and the suggested ranges for paragraph 1 and Table 4 (~1.4–1.9 B USD bottom line).